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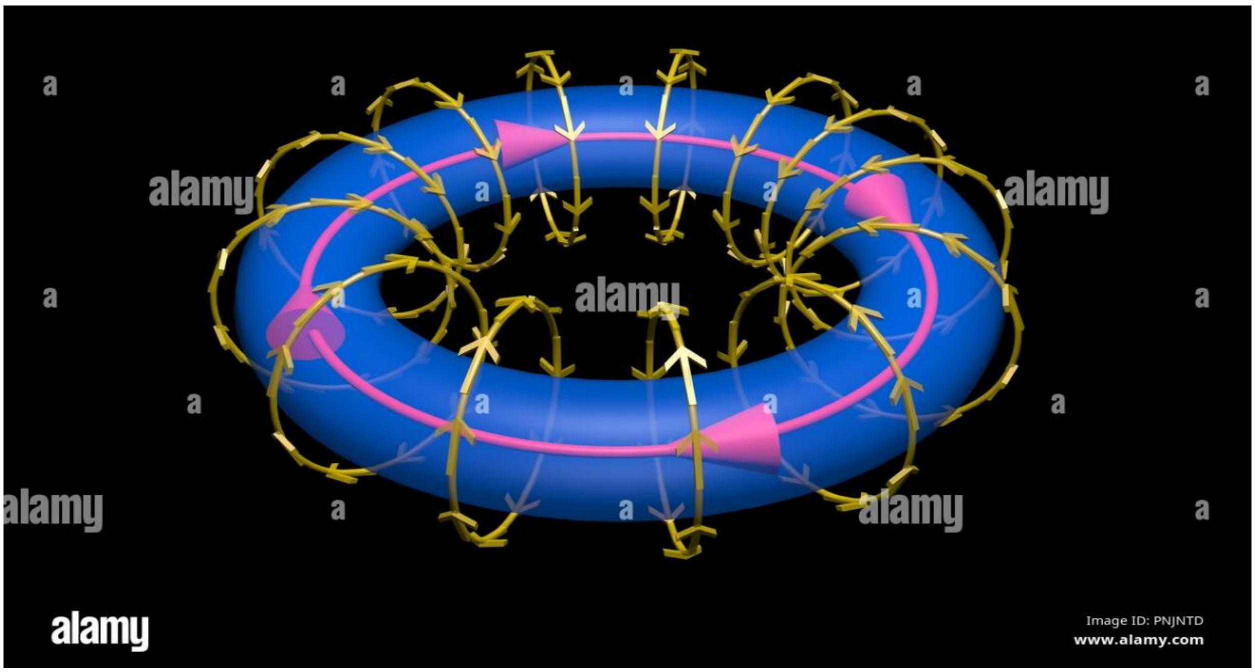
Toroid

❖ Definition:

A toroid is a hollow circular ring, shaped like a doughnut, made by bending a solenoid (coil of wire) into a circular form. It consists of many closely wound turns of wire, with negligible space between any two turns.

Toroids are widely used in applications such as transformers, inductors, and magnetic storage devices because they confine the magnetic field within their core, minimizing electromagnetic interference. In general, toroids act as inductors in electric circuits and are particularly suited for applications requiring high frequencies.

❖ Diagram:



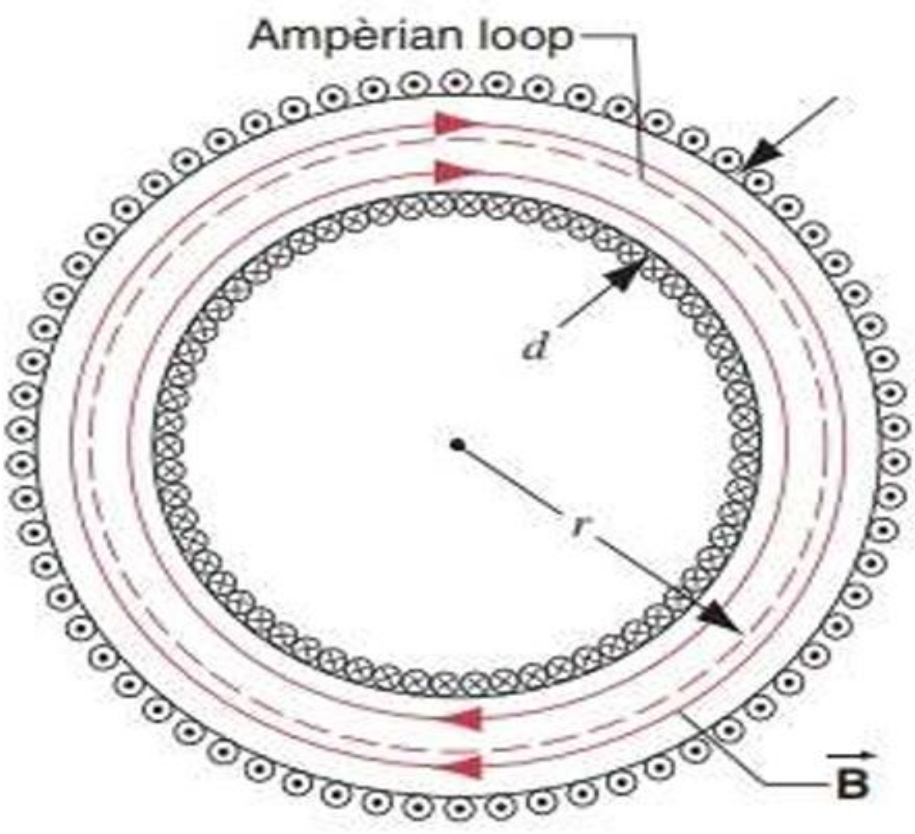
Magnetic field due to Toroid

❖ Background:

The study of magnetic fields generated by currents began with **Oersted's discovery** in 1820 that a current-carrying wire creates a magnetic field. Later, **Ampère's Circuital Law** provided a mathematical framework for calculating magnetic fields due to symmetric current distributions, such as those in solenoids and toroids. The toroid, a solenoid bent into a circular shape, was developed to confine magnetic fields and reduce losses due to leakage.

❖ Diagram:

The Diagram of the magnetic field due to toroid is given below:



❖ Explanation:

1. Structure

- Consider a **hollow circular ring** with many closely wound turns of a current-carrying wire around it. The wire carries a current "I", creating a magnetic field "B" inside the toroid.
- The figure shows the cross-sectional view of the toroid, with current flowing into and out of the plane alternately, represented by dots (out of the plane) and crosses (into the plane).

2. Ampèrian Loop

- The loop shown in the figure is an **Ampèrian loop**. A closed circular path passes through point P inside the toroid.

- The Ampèrian loop forms circles around the central axis of the toroid, ensuring that the magnetic field inside the toroid is circular.

3. *Magnetic Field (B)*

- Inside the toroid, the magnetic field is circular and tangential to the Ampèrian loop at every point. The field strength depends on the radius r.
- Outside the toroid (both inside the hollow region and outside the toroid itself), the magnetic field is negligible due to symmetry and cancellation of contributions.
- Due to the symmetric field, at any point on the Ampèrian loop, the magnetic field has the same magnitude.

❖ Derivation:

➢ We know that "Ampère's law" is given by:

$$\oint B \cdot dl = \mu_0 I_{enclosed} \quad \text{.......................(1)}$$

❖ *Here:*

- *B: Magnetic field at a point on the loop.*
- *dl: Differential length element along the Ampèrian loop.*
- *$I_{enclosed}$: Total current enclosed by the loop.*
- *μ_0: Permeability of free space.*

➢ *By taking the integration of the "$\oint B \cdot dl$" we get:*

$$\oint B \cdot dl = B \cdot (2\pi r) \quad \text{....................................(a)}$$

➢ *The total current enclosed by the loop is:*

$$I_{enclosed} = n I \quad \text{.......................................(b)}$$

❖ *where:*

- *$n = N / (2\pi R)$ is the number of turns per unit length of the toroid.*
- *N: Total number of turns.*
- *R: Average radius of the toroid.*
- *I: Current through each turn of the wire.*

➢ *Now by putting the values of eq (a) and (b) in eq (1), we get:*

$$B \cdot (2\pi r) = \mu_0 n I$$

➢ *Now by rearranging the given above eq, we get:*

$$B = \mu_0 n I / (2\pi r)$$

This is the final form of the Eq for the magnetic field due to the toroid.

Special Cases

❖ *Magnetic Field inside the Toroid*

- The eq of the magnetic field inside the toroid Is as follows:

$$B = \mu_0 n I/(2\pi r)$$

❖ *Magnetic Field outsidce the Toroid*

- Outside the toroid, the currents in adjacent turns of the toroid produce opposing magnetic fields which cancel each other out, thus:

$$B = 0$$

❖ *Magnetic Field inside an non-ideal Toroids*

- The eq of the magnetic field inside the toroid Is as follows:

$$B=\mu_0 \mu_r n I/(2\pi r)$$

For toroids with a ferromagnetic core, the field is enhanced by the material's relative permeability "μ_r"

❖ *Magnetic Field outside an non-ideal Toroids*

- In a non-ideal case, the magnetic field outside the toroid is difficult to calculate directly due to its dependence on irregularities. However, using the Biot-Savart law we get:

$$B=\mu_0 I/(4\pi) \int (dl\times \hat{r}) /r^2$$

and by computeing the contribution of each segment of the current-carrying wire we will get our resulted magnetic field.